

BungVision

The Bunsen valve check before the battery wash. How to run the station, read a PASS or FAIL, clear a stopped line, and check your production numbers.

For machine operators

Read start to finish once

Keep near the station

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1 What BungVision Does

BungVision is a camera that watches each battery on its way to the **battery wash**. It looks at the top of the battery and checks that every

Bunsen valve is in place before the battery gets washed. A Bunsen valve is the small round plug that seals a cell on top of the battery.

The Bunsen valves are what keep wash water out of the battery. If one is missing, or is not pushed in all the way, water can get inside during the wash. Water inside the battery ruins it. So the job here is simple: **catch a Bunsen valve that is missing or not in place, before the battery reaches the wash.**

BungVision uses a trained model (its "eyes and brain") to find the battery and count the Bunsen valves many times per second. If a Bunsen valve is not in place, it shows a big red **REJECT LATCHED** message and tells the machine (the PLC) to stop the line, so the Bunsen valve can be put in before the battery goes any further.

IN PLAIN WORDS

BungVision = an automatic set of eyes at the wash. It makes sure every Bunsen valve is in place, so water cannot get into a cell and ruin a battery.

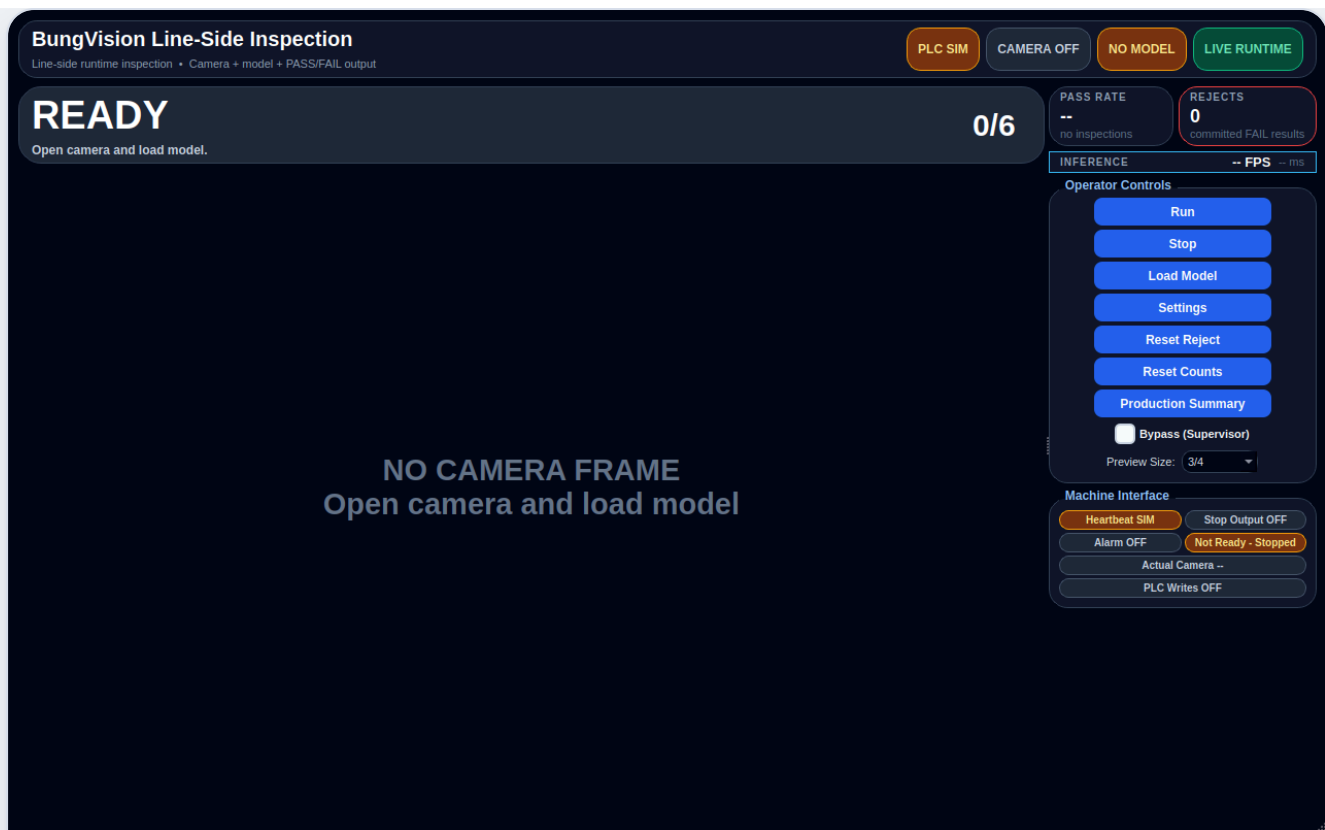
A STOP IS A SAVE, NOT A SCRAP

When the line stops, the battery is usually still fine — that is the point of catching it here. Put the Bunsen valve in place and it can go on to the wash. If it had gone through with a Bunsen valve missing, it would have been ruined.

2

The Main Screen

This is the screen you will use all shift. The big panel on the left is the live camera and the result. The column on the right has your buttons and the machine status.



The main screen at startup. Left: the camera view and the big result panel. Right: your buttons (top) and the machine status lights (bottom).

The parts of the screen

- **Result panel** (big box, top left) — shows **PASS**, **FAIL**, or **WAIT** for the battery being checked, and the Bunsen valve count like **6/6**.
- **Camera view** (below the result) — the live picture from the camera.
- **Status chips** (top right corner) — quick health of the **PLC**, **Camera**, and **Model**.
- **Metric cards** — **Pass Rate**, **Rejects** this session, and **Inference** (how fast the camera is thinking, in frames per second).
- **Operator Controls** — the buttons you press: Run, Stop, Load Model, Settings, Reset Reject, Reset Counts, Production Summary.
- **Machine Interface** — the lights that show what BungVision is telling the machine (heartbeat, stop request, alarm, ready).

3 Starting and Stopping

Before the line runs, BungVision needs its **model** loaded and its **camera** open.

- 1 **Load the model.** Press **Load Model**. When it is ready, the **MODEL** chip at the top turns green. The model is the file that knows what a battery and a Bunsen valve look like.
- 2 **Start the camera.** Press **Run**. The camera view goes live and the **CAMERA** chip turns green.
- 3 **Watch the result panel.** As batteries pass, you will see PASS, FAIL, or WAIT for each one.
- 4 **To stop, press **Stop**.** The camera stops and BungVision tells the machine it is no longer running.

IF THE CAMERA FREEZES

Press **Run** again. That closes and re-opens the camera. If it keeps stopping after about a minute, tell maintenance — the camera's USB power may be turning off (there is a one-time fix for this on the Jetson).

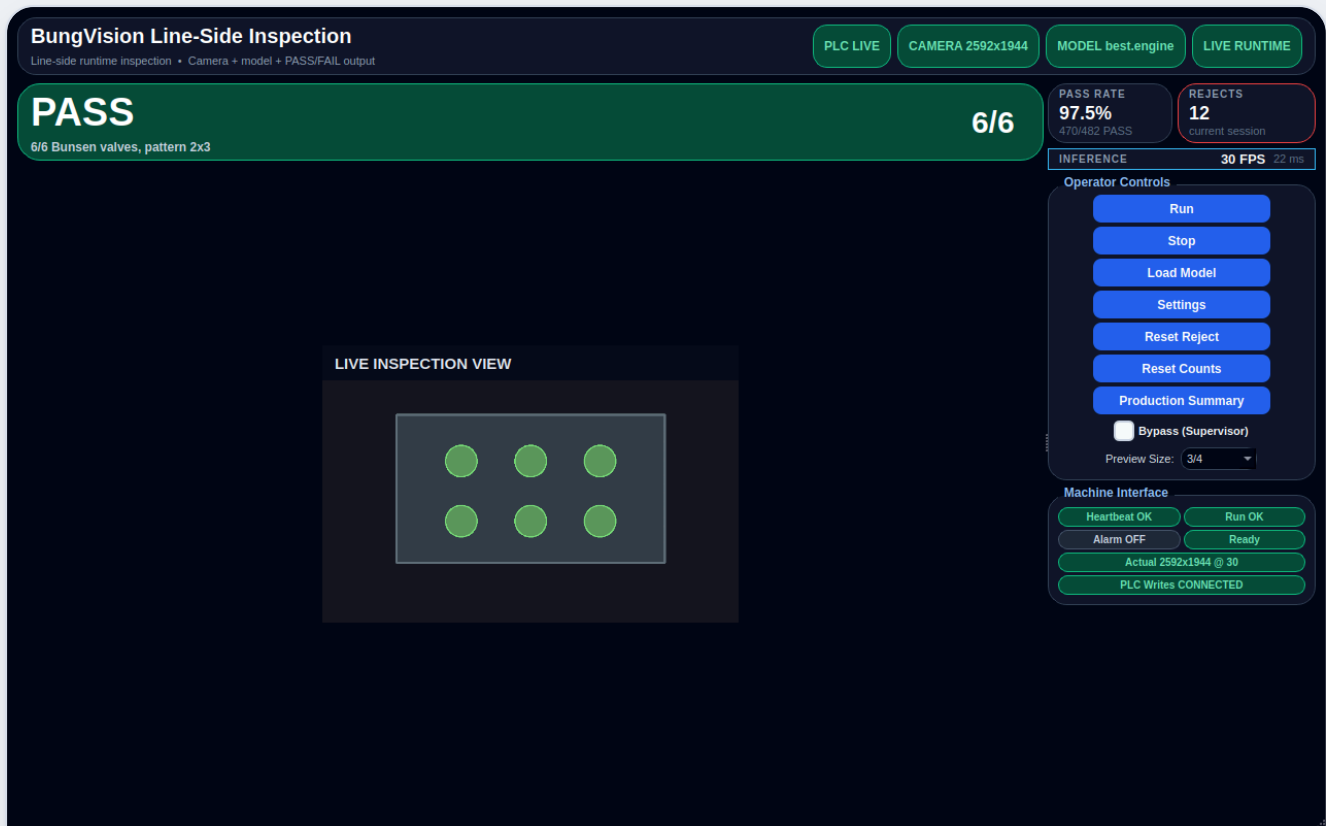
4 How PASS and FAIL Are Decided

For every battery, BungVision does the same simple check:

- Find the battery in the picture.
- Count the Bunsen valves that belong to that battery.
- Compare the count to the number it should have (normally **6**).

A PASS

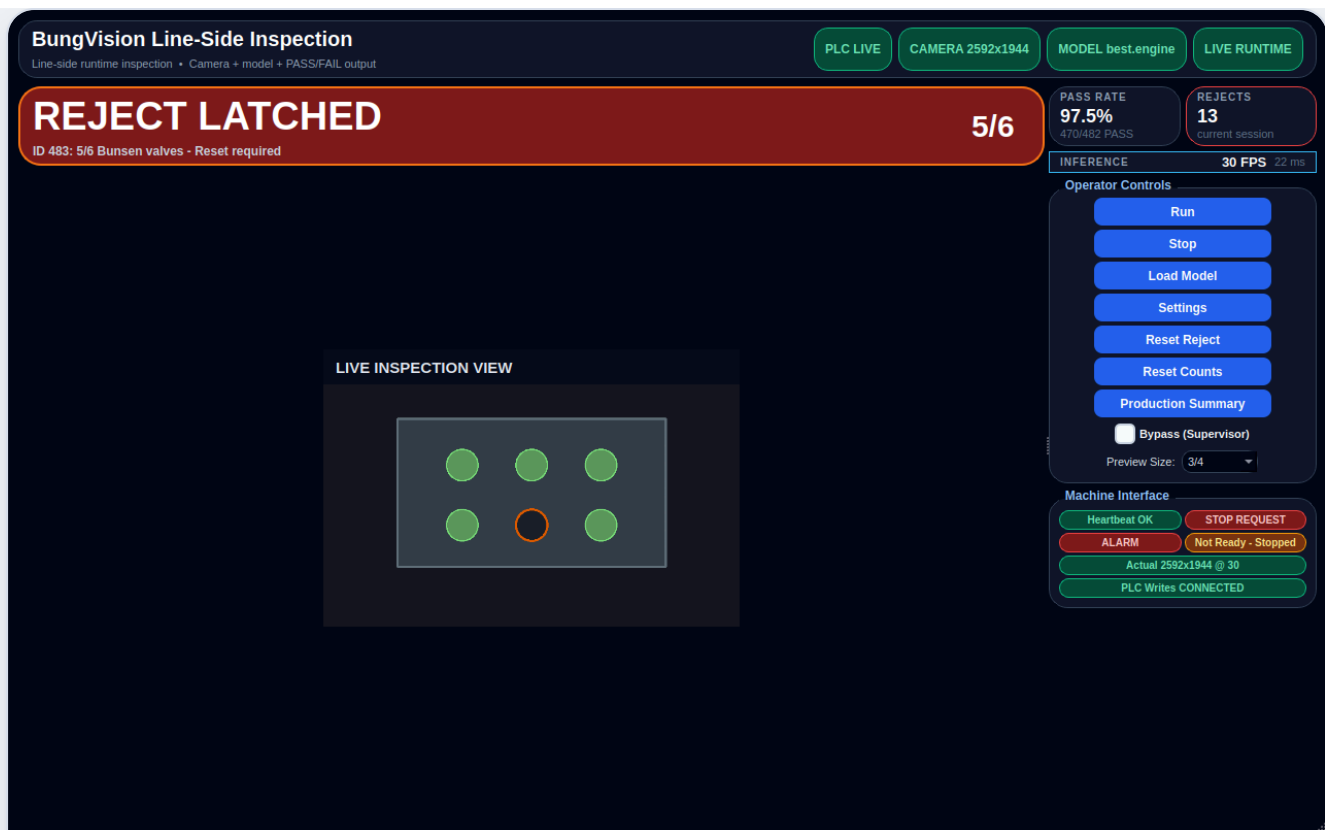
All 6 Bunsen valves are in place and in the right pattern. The battery is closed up and safe to wash, so the panel turns green.



A PASS. All six Bunsen valves are found. The panel is green, it reads `6/6`, and the reason says `6/6 Bunsen valves, pattern 2x3`.

A FAIL

A Bunsen valve is missing, or one is not sitting where it should be. The battery is **not** closed up for the wash, so the panel turns red and the line stops before it gets there.



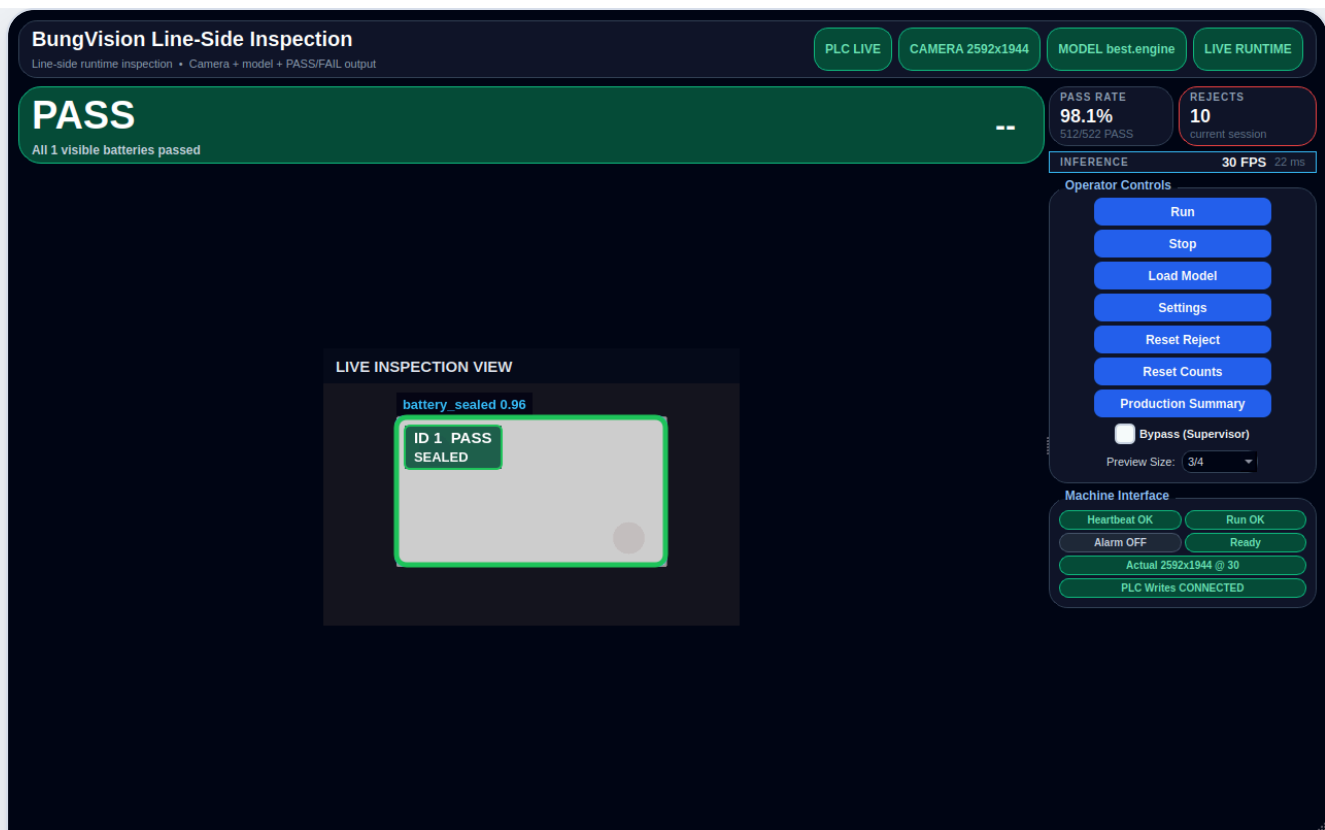
A FAIL. One Bunsen valve is missing, so the count is **5/6**. The panel turns red and locks on **REJECT LATCHED**.

Sealed batteries

Some battery types have **no Bunsen valves at all** — they are sealed at the factory. There is nothing to count on those, so BungVision skips the check for them and passes them straight through.

You can always tell when this happened. The badge on the battery reads

SEALED instead of a count, and the big number shows **--** instead of **6/6**.



A sealed battery. No Bunsen valves, and none needed. The badge reads **SEALED**, the count shows --, and the battery passes.

HOW BUNGVISION KNOWS

It goes by the battery type the camera recognizes. Sealed types are taught to the model with the word `sealed` in their name. Every other battery — including any type BungVision does not recognize — still gets the full Bunsen valve check.

IF YOU SEE SEALED ON A BATTERY THAT NEEDS BUNSEN VALVES

Stop the line and tell a supervisor. That means the camera has mistaken the battery type, and a battery that needs Bunsen valves would go to the wash unchecked. Do not just keep running.

What "WAIT" means

Sometimes you will see `WAIT`. That is normal. It means the battery is not all the way in view yet — it is still coming in from the edge of the

picture. BungVision holds off on judging until it can see the whole battery. It will not stop the line just because a battery is half in frame.

The extra checks

A few settings make the check smarter. A supervisor sets these on the **Inspection** tab.

Setting	Value
Bunsen Valves Expected Bunsen valve count per battery before a PASS can commit.	6
Entry Grace WAIT frames after a new battery appears so Bunsen valves can enter view.	12
Match px Maximum center movement allowed when matching a battery track.	180
Locked IoU % Overlap required to keep a committed PASS/FAIL locked while visible.	25
Full View Margin % Battery must be this far from the frame edge before PASS/FAIL grading can commit.	3
Pattern Tol % How far off the fitted row/grid a valve may sit and still be counted. Higher = looser (more forgiving of skew, but off-pattern items are more likely to count).	25
Debounce Frames a PASS/FAIL must stay stable before it is counted.	6
Clear Frames Frames with no match before a tracked battery is cleared.	10
Track IoU % Overlap threshold used to keep moving batteries tied to the same ID.	5
☒ Require full view before grading Hold a partial infeed/edge battery in WAIT instead of grading it FAIL before the full lid is visible.	
☒ Validate Bunsen valve pattern Auto-detect the single-row or 2x3 layout and count only Bunsen valves that sit on it. Off-pattern detections (terminals, mounting holes, strays) are ignored so they cannot inflate the count. Row spacing is not constrained.	

Buttons: Apply, Save, Save + Close, Load Model, Cancel

Inspection settings. **Bunsen Valves** is how many each battery must have (6). **Require full view** turns on the WAIT behavior. **Validate Bunsen valve pattern** makes the count ignore anything off the pattern.

RULE	WHAT IT DOES
Bunsen Valves	How many Bunsen valves must be in place before the battery can go to the wash. Default is 6.
Require full view before grading	Holds a battery in WAIT until the whole battery is on screen, so an edge battery does not stop the line by mistake.
Validate Bunsen valve pattern	Finds the row (or 2×3 grid) the Bunsen valves sit in, then counts only the ones on that pattern . Anything off the pattern — a battery post, a mounting hole, a stray — is ignored and does not add to the count. Uneven spacing along the row is fine.
Pattern Tol %	How far off the row or grid a Bunsen valve may sit and still be counted. Higher is more forgiving.

WHAT A PASS NEEDS

The whole battery in view **and** 6 Bunsen valves sitting on the pattern — the battery is closed up and cleared for the wash. BungVision works out the layout by itself — a single row or a 2×3 grid — so both battery styles are handled with no setup.

ONE THING THIS CANNOT CATCH

The pattern check only removes Bunsen valves that are **off** the pattern. If something is wrongly seen as a Bunsen valve right where a real Bunsen valve belongs (for example a deep empty hole), it is on the pattern and still counts. That kind of mistake has to be fixed by retraining the model, not by this setting.

5

When a Battery Is Rejected

The moment BungVision commits a FAIL, it does two things at once:

- 1 It counts the fail. Your **Rejects** number goes up by one. That number tells you how many batteries the system stopped before the wash.
- 2 It **latches the reject**. "Latch" means it locks on and stays on. The panel shows **REJECT LATCHED** and BungVision turns on **STOP REQUEST** and **ALARM** to the machine so the line stops.

The reject stays locked until someone clears it. This is on purpose — it makes sure a person looks at the battery before it can go on to the wash.

LOOK AT THE MACHINE INTERFACE LIGHTS

On a reject you will see **STOP REQUEST** and **ALARM** turn red, and **Ready** changes to **Not Ready - Stopped**. That is BungVision telling the machine to hold.

6 Clearing a Reject (Reset Reject)

When the line stops for a reject, here is what you do:

- 1 **Fix the battery.** Put the missing Bunsen valve in, or push in the one that is not all the way down. If you cannot fix it on the line, pull the battery off.
- 2 **Press** **Reset Reject**. This unlocks the latch. BungVision sends a reset to the machine and goes back to **READY**.
- 3 **Let it re-inspect.** BungVision will look at the fixed battery again. It has to see all 6 Bunsen valves in place before the line keeps running.

Reset Reject only clears the lock. It does **not** erase the fail from your count. That is on purpose — you still want to know how many

batteries the system caught before the wash this shift.

THE MACHINE CAN RESET IT TOO

Your PLC can clear the reject on its own by turning on the `BungVision_ResetRequest` tag. BungVision watches that tag and, when it turns on, does the same thing as pressing Reset Reject. Ask a supervisor if your line uses this.

7 Reset Reject vs Reset Counts

These two buttons look alike but do very different jobs. Do not mix them up.

BUTTON	WHAT IT DOES	WHEN TO USE IT
<code>Reset Reject</code>	Unlocks the current reject so the line can run again. Your counts stay the same.	Every time the line stops for a missing Bunsen valve, after you put it in place.
<code>Reset Counts</code>	Sets the Parts, Pass, and Reject counters back to <code>0</code> and starts a fresh session. It also clears any active reject.	At the start of a shift or when you change to a new product.

EASY WAY TO REMEMBER

Reset Reject = "let me keep going." **Reset Counts** = "start my numbers over." Use Reset Counts only at shift start, never just to clear a stop.

8 The Heartbeat and the PLC

The **PLC** is the machine's controller. BungVision talks to it over the network. A supervisor sets this up on the **PLC** tab, but you should

understand the lights.

The screenshot shows the 'PLC' tab in the BungVision interface. It contains the following elements:

- PLC IP:** Controller address for pylogix writes. Value: 192.168.1.10
- Heartbeat ms:** How often the heartbeat bit toggles. Value: 500
- Enable Writes:** A checkbox that is currently unchecked. Description: Turn on only after tags are verified.
- Vision runtime running / active:** TRUE when BungVision is actively running inspection or when Bypass is active. Tag: BungVision_Running
- Conveyor stop request:** TRUE when BungVision has a committed reject/fault and Bypass is not active. Tag: BungVision_StopRequest
- Vision ready / permissive:** TRUE only when BungVision is genuinely able to inspect: runtime running, camera open, recent good frame, model loaded, prediction not faulted, and no stop request. Tag: BungVision_Ready
- Reset acknowledgement pulse:** Momentary TRUE pulse when Reset is pressed in BungVision. Tag: BungVision_Reset
- Vision bypass active:** TRUE when the operator has enabled Bypass. Tag: BungVision_Bypass
- Vision alarm active:** TRUE when BungVision wants the operator/ PLC to treat the condition as an alarm. Tag: BungVision_Alarm
- Communication heartbeat:** BOOL that toggles while PLC writes are enabled. Tag: BungVision_Heartbeat
- Reset request (PLC -> Vision input):** BOOL INPUT written by the PLC. Tag: BungVision_ResetRequest

At the bottom of the PLC tab are three buttons: Test Conn, Test HB, and Test Stop. Below the PLC tab are five buttons: Apply, Save, Save + Close, Load Model, and Cancel.

PLC settings. Each line is a "tag" — a named signal shared with the machine. The heartbeat, ready, stop, and reset tags live here.

What the heartbeat is

The **heartbeat** is a signal that turns on and off over and over, like a pulse, the whole time BungVision is alive and talking to the PLC. The machine watches this pulse. As long as it keeps beating, the machine knows BungVision is awake and working.

If the heartbeat **stops** beating, the machine knows something is wrong with BungVision (it froze, lost power, or lost the network) and can stop the line for safety. Think of it like a person nodding "I'm still here" every half second.

The Machine Interface lights

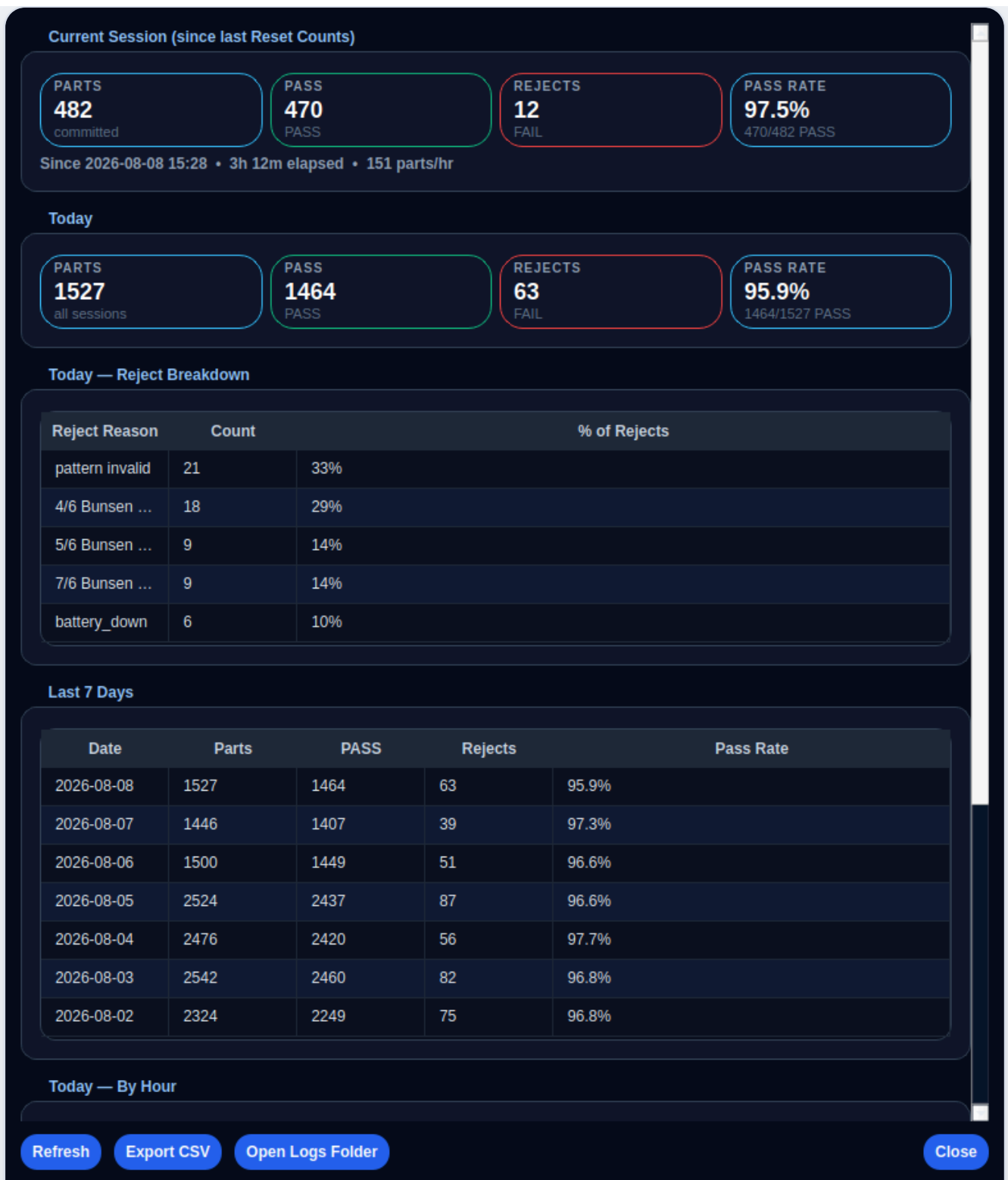
LIGHT	MEANING
Heartbeat	The pulse that proves BungVision is alive. It should keep changing.
Stop / Run	Green when running normally; red STOP REQUEST when a reject wants the line stopped.
Alarm	Turns on with a reject, for the stack light or horn.
Ready	Green only when everything is healthy: running, camera open, model loaded, and no stop. Not green means BungVision is not ready to inspect.
Actual Camera	The real camera size and speed, like <code>2592x1944 @ 30</code> .
PLC Writes	Shows if BungVision is really connected to the PLC.

Bypass (supervisor only)

The **Bypass (Supervisor)** check box lets a supervisor keep the line moving even when BungVision would ask it to stop. It is for setup and troubleshooting only. Bypass does **not** make the machine "Ready" — it only stops BungVision from sending the stop. Use it with care.

9 Production Tracking

Press **Production Summary** to see how the line is doing. This screen only shows numbers — you cannot break anything here.



Production Summary. Your current session at the top, then all of today, a breakdown of why batteries failed, and the last 7 days.

- **Current Session** — Parts, Pass, Rejects, and Pass Rate since the last **Reset Counts**. It also shows parts per hour.
- **Today** — the same numbers for the whole day, across every session.

- **Today — Reject Breakdown** — the reasons batteries failed (like 5/6 Bunsen valves), so you can see the most common problem.
- **Last 7 Days** — one row per day, with the pass rate for each.
- **Today — By Hour** — parts and rejects broken down by hour (scroll down to see it).

SAVE THE NUMBERS

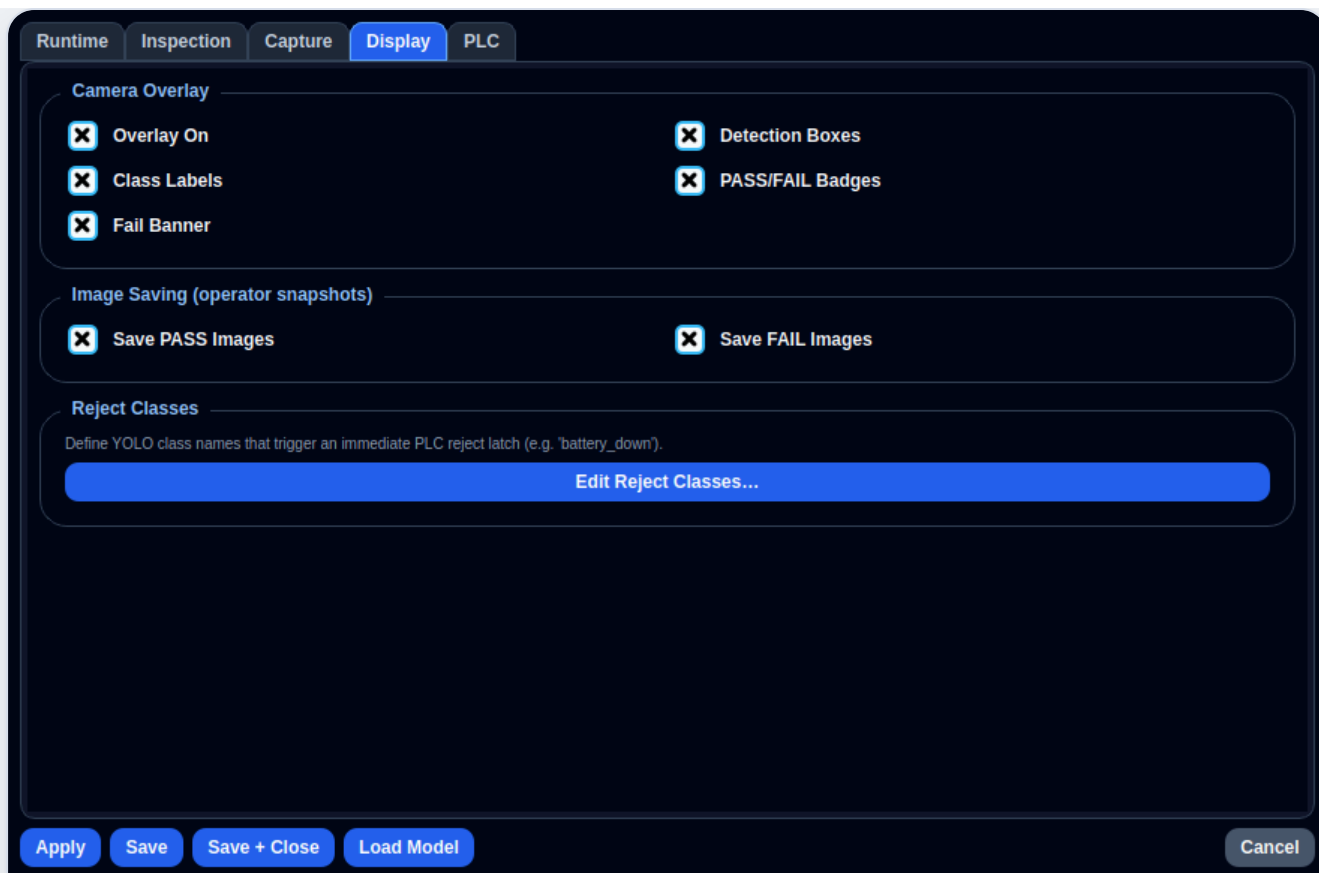
Press Export CSV to save a spreadsheet of the history, or Open Logs Folder to find the saved files. This history stays even after BungVision restarts.

10 Settings

Press Settings to open the setup window. It has tabs across the top. Most operators never need to change these — a supervisor sets them.

The screenshot shows the 'Runtime' tab of the BungVision Settings window. The window has a dark theme and a tabbed interface at the top with 'Runtime', 'Inspection', 'Capture', 'Display', and 'PLC'. The 'Runtime' tab is active. The settings are organized into two columns. The left column includes: 'Backend' (OpenCV), 'OpenCV API' (Auto), 'Cam FPS' (30), 'Height' (1944), 'Manual us' (5000), 'Enable Basler ROI' (checkbox), 'ROI Offset X' (0), 'ROI Width' (2592), 'Model Task' (Auto), and 'Class Names'. The right column includes: 'OpenCV Src' (0), 'Basler SN' (blank), 'Width' (2592), 'Exposure' (Manual), 'Gain' (0), 'ROI Offset Y' (0), 'ROI Height' (1944), 'Device' (blank), and 'Image Size'. At the bottom, there are buttons for 'Apply', 'Save', 'Save + Close', 'Load Model', and 'Cancel'.

Runtime tab. Which camera to use, its size and speed, and which model file to load.



Display tab. Turn the camera overlay boxes on or off, choose whether to save PASS and FAIL pictures, and open the reject class list.

The **Reject Classes** list lets you name shapes that should stop the line right away — for example a tipped-over battery. If the camera sees that shape, it rejects at once, without counting Bunsen valves.

Add YOLO class names below. When any detection matches one of these labels the PLC reject latch fires immediately, regardless of Bunsen valve count. Names are case-insensitive. Example: **battery_down**

battery_down
label_missing

e.g. battery_down

Add

Remove Selected

OK

Cancel

Reject Classes. Type a class name (like `battery_down`) to make it an instant reject.

11 Collecting Training Pictures

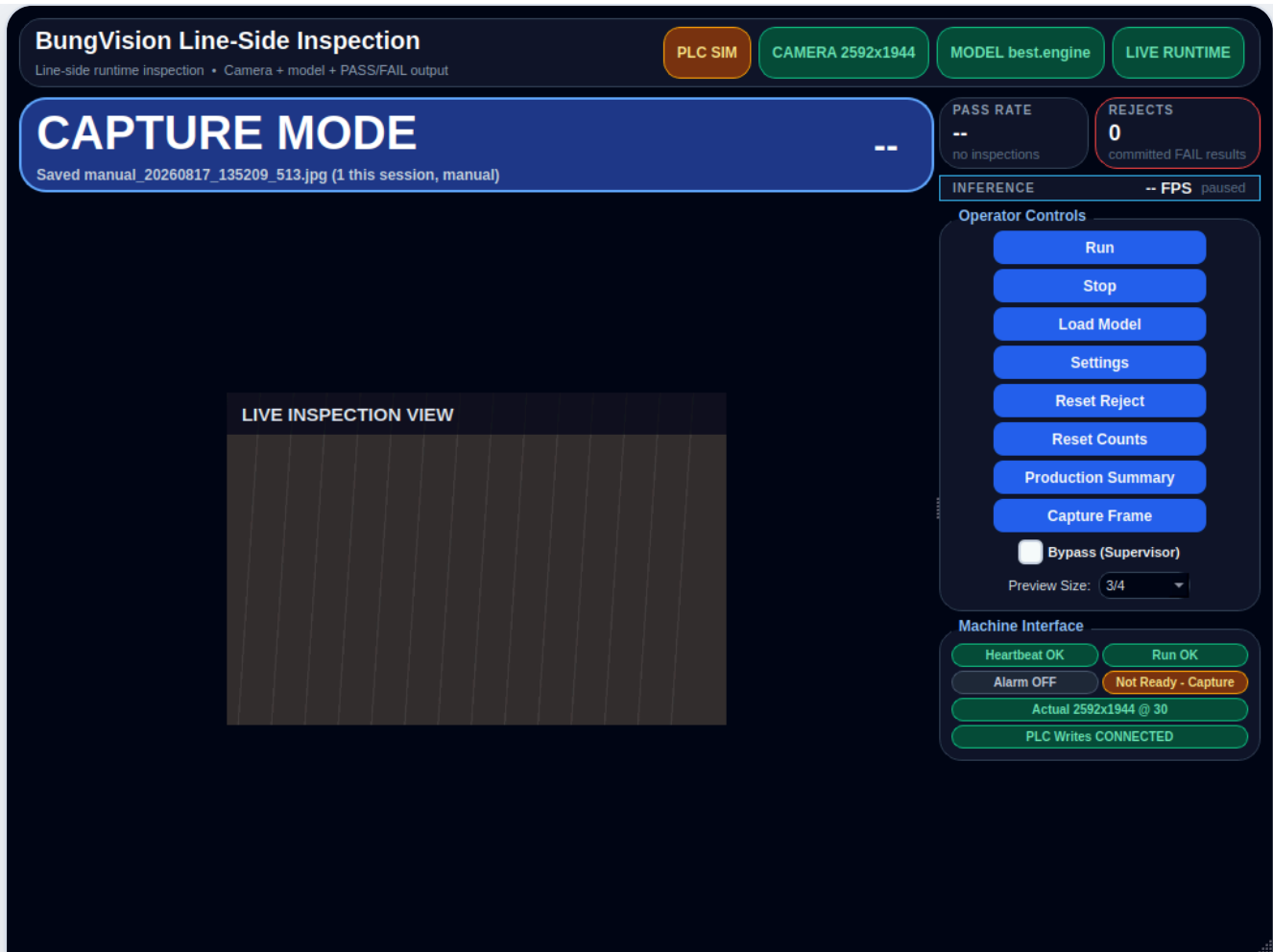
The camera learns from pictures. To teach it something new — an empty belt, a battery placed by hand, a tricky Bunsen valve that keeps fooling it — BungVision has a **capture mode** that saves plain pictures for training.

INSPECTION IS OFF IN CAPTURE MODE

While capture mode is on, BungVision is **not inspecting anything**. It will not catch a missing Bunsen valve and it tells the machine it is **not ready**. Only use it when the line is stopped for this purpose, and turn it off when you are done.

How to use it

- 1 Press **Settings** and go to the **Capture** tab.
- 2 Tick **Manual capture mode (inference OFF)**, then press **Save + Close**.
- 3 The screen turns blue and reads **CAPTURE MODE**. A **Capture Frame** button appears in the buttons list.
- 4 Set up your shot — an empty belt, or a battery placed where you want it — and press **Capture Frame** (or tap the **space bar**). The line under the title tells you the file name and how many you have taken.
- 5 When you are finished, go back to **Settings → Capture**, untick the box and save. Inspection starts again.



Capture mode. The blue panel means inspection is paused. The **Capture Frame** button saves the picture; the line below the title confirms each save.

Runtime
Inspection
Capture
Display
PLC

☒ **Save PASS examples**
Periodically save passing examples for retraining review.

☒ **Save FAIL examples**
Save failed inspections immediately for review and correction.

☒ **Save annotated images**
Also save an overlay image with OBB/box detections drawn.

☒ **Save detection JSON**
Write label-tool JSON with OBB shapes/corners when available.

☐ **Save YOLO OBB .txt candidate**
Write YOLO OBB corner labels when OBB detections are available; otherwise box labels.

PASS sample: 1 /
Save one PASS image every N committed PASS results.

Open Captures

Open Fail

Open Pass

☒ **Manual capture mode (inference OFF)**
Keep the camera live but pause inference, and show a Capture Frame button on the main screen (shortcut: Space). Raw unannotated frames are saved to manual_captures/ for training. Nothing is graded and PLC Ready stays off while this is on.

Auto every N sec
0 = manual only. Above 0, capture a frame automatically on that interval (useful for blank-conveyor footage).

Open Manual Captures

Review/correct saved examples in the labeling tool before retraining.

Apply
Save
Save + Close
Load Model
Cancel

Capture tab. Tick **Manual capture mode** to start. **Auto every N sec** takes pictures on its own — handy for filming an empty belt. Leave it at to take them one at a time.

Pictures are saved in the **manual_captures** folder, on their own — they are not mixed in with your PASS and FAIL pictures. Use **Open Manual Captures** on the Capture tab to find them.

WHAT MAKES A USEFUL PICTURE

Take the ones the camera gets wrong. If it keeps seeing a Bunsen valve that is not there, photograph that exact battery from a few angles and in the lighting you really run. A handful of the hard ones is worth more than a hundred easy ones.

Everyday Tasks

YOU WANT TO...	DO THIS
Start inspecting	Press Load Model , then Run .
Stop inspecting	Press Stop .
Clear a stopped line after a reject	Fix the battery, then press Reset Reject .
Start a new shift's numbers	Press Reset Counts .
See how the shift is going	Press Production Summary .
Un-freeze the camera	Press Run again.
Take pictures for training	Settings → Capture → tick Manual capture mode , then use Capture Frame (or the space bar). Untick it when done.

Fixing Common Problems

PROBLEM	WHAT IS LIKELY WRONG	WHAT TO TRY
Camera stops on its own after about a minute	The camera's USB power is turning off	Press Run to restart. Tell maintenance so they can apply the one-time USB power fix.
Picture is very dark	Camera exposure is set too low	Tell a supervisor to check the exposure setting for this camera.
Frames per second look low	The camera decode is slow	Supervisor: set the camera to the GStreamer option so the hardware helps decode.
Good batteries show FAIL	Pattern check is too strict, or a Bunsen valve is dirty/hidden	Clean the battery top. If it keeps happening, a supervisor can raise the Pattern Tolerance.
Machine will not run even on PASS	PLC not connected, or not "Ready"	Check the PLC Writes and Ready lights. If not green, get a supervisor.
Machine lights flicker on and off by themselves (Bypass, Ready, Stop)	A second copy of BungVision is open, and both are talking to the machine at once	Only one copy may run at a time. Close the extra copy. BungVision now refuses to open a second copy and will tell you one is already running.
Says TAG CONFLICT on the PLC Writes light	Two machine signals were set to the same tag name	Get a supervisor: in Settings → PLC, every signal needs its own tag name.
Says model is not loaded	The model file did not load	Press Load Model . If it still fails, get a supervisor.
A battery passed with no Bunsen valves in it	The camera read it as a sealed type, which needs none	Check the badge. If it says SEALED on a battery that really needs Bunsen valves, stop the line and get a supervisor — the type is being read wrong.

Screen is blue and says **CAPTURE MODE**

Capture mode was left on, so nothing is being inspected

Settings → Capture → untick **Manual capture mode** and save.

WORD	MEANING
Bunsen valve	The round plug that seals a cell on top of the battery. It keeps wash water out of the cell. BungVision counts these.
Battery wash	The wash stage the batteries go to next. BungVision is the check just before it.
Cell	One of the six chambers inside the battery. Each one is sealed by a Bunsen valve. If water gets into a cell, the battery is ruined.
Model	The trained file that knows how a battery and a Bunsen valve look. BungVision needs it loaded to inspect.
PLC	The machine's controller. BungVision tells it when to stop and go.
Heartbeat	A signal that pulses on and off to prove BungVision is alive and talking to the PLC.
Latch	To lock on and stay on. A reject latches so no battery reaches the wash with a Bunsen valve out of place.
WAIT	The battery is not fully in view yet, so BungVision holds off judging it.
Inference	The camera "thinking" — how fast it checks frames, shown as frames per second (FPS).
Bypass	A supervisor switch that lets the line run even when BungVision would stop it. For setup only.
Sealed battery	A battery type built with no Bunsen valves. BungVision skips the Bunsen valve check on these and shows a SEALED badge.
Capture mode	A setting that pauses inspection so you can save plain pictures for training. Nothing is checked while it is on.
Session	Your run of counting since the last Reset Counts.

BungVision Line-Side Inspection · Operator Manual. Screens shown with sample data for teaching; your live screen will show your camera and your real numbers. Keep this guide at the station.